Complete Technical Reference & Pipeline Specification across 7 Operational Sectors



⏱️ Computational Timing, Hardware Benchmarks & Performance Metrics

Measured and verified on an **NVIDIA GeForce RTX 4050 Laptop GPU (6GB VRAM, CUDA 13.2)** with an Intel 14-thread CPU host architecture.

1. OFFLINE RECONSTRUCTION & TRAINING PIPELINE (TOTAL: ~30 – 40 MINS)

SECTOR / PHASE	TASK DESCRIPTION	MEASURED DURATION	HARDWARE RESOURCE
Sector 1: Video Curation	FFmpeg 5 FPS extraction + dynamic person purge (frames 222–263) + blur filter	`1m 45s (~1.5 – 2 mins)`	Multi-core CPU & Storage I/O
Sector 2: SfM Tracking	COLMAP SIFT extraction, exhaustive matching & bundle adjustment (370 cameras, 89,839 pts)	`15m 12s (~12 – 18 mins)`	14 CPU Threads
Sector 3: 3DGS Training	Brush CUDA 30,000 steps with SuperClean scale penalty (5e-8), growth stop at 6k	`17m 30s (~15 – 20 mins)`	RTX 4050 GPU (~30 iter/s, ~3.8GB VRAM)
Sector 4: Filtering Pass	KD-Tree SOR (k=16, 1.5σ) + needle aspect clamping + viewpoint SH zeroing	`1m 10s (~1 – 1.5 mins)`	CPU / NumPy Vectorization
Sector 5: Manifest Packaging	Rodrigues axis rotation derivation & scene.json generation	`~5 seconds`	Python / JSON I/O
Total End-to-End Pipeline	**From raw 1080p walkthrough video to production-ready 3D Gaussian Splat**	`~35m 32s (30 – 40 mins)`	**Hybrid GPU / CPU Acceleration**

2. REAL-TIME CLIENT-SIDE VIEWER PERFORMANCE (BROWSER WEBGL)

PERFORMANCE METRIC	MEASURED VALUE	TECHNICAL MECHANISM
Initial Model Download & Parse	`1.5 – 2.5 seconds`	Ingestion of 24.3 MB binary `scene_superclean.ply` (102,242 splats)
Web Worker Radix Depth Sorting	`5 – 8 milliseconds`	Off-main-thread multi-threaded sorting via `SharedArrayBuffer`

PERFORMANCE METRIC	MEASURED VALUE	TECHNICAL MECHANISM
Per-Frame GPU Render Latency	`6.0 – 12.0 milliseconds`	Three.js WebGL tile-based rasterization
Interactive Frame Rate (FPS)	`83 – 144+ FPS`	Fluid desktop free-roaming and orbit manipulation
Client Memory Utilization	`~210 MB VRAM / ~340 MB RAM`	Compact representation (27 floats per Gaussian at SH=2)

`reconstruction/preprocessing/extract_curate_v3.py`

PRIMARY TOOL	INPUT ASSET
FFmpeg 9.0.2 & OpenCV Laplacian	IMG_3363.MOV (1080p, 30 FPS Walkthrough)
OUTPUT ASSET	**MEASURED DURATION**
data/processed/custom-video-v3/images/ (370 frames)	1m 45s (~1.5 – 2 mins)

Algorithmic Mechanism:

- **Temporal Resampling:** Decodes stream at 5.0 FPS into 1280×720 frames using Lanczos scaling.
- **Dynamic Artifact Removal:** Purges frames 222–263 to remove transient human motion and prevent phantom floaters.
- **Laplacian Variance Scoring:** Evaluates edge variance $\mathrm{Var}(\nabla^2 I)$ and discards the blurriest 25% of panning frames to yield 370 curated frames.

```python
def curate():
    all_raw = sorted(glob.glob(os.path.join(raw_dir, "frame_*.jpg")))
    person_frames = {f"frame_{i:05d}.jpg" for i in range(222, 264)}
    non_person_files = [f for f in all_raw if os.path.basename(f) not in person_frames]

    scored_frames = []
    for fpath in non_person_files:
        img = cv2.imread(fpath)
        gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
        lap_var = float(cv2.Laplacian(gray, cv2.CV_64F).var())
        scored_frames.append((fpath, lap_var))

    scored_frames.sort(key=lambda x: x[1])
    k_drop = int(len(scored_frames) * 0.25)
    kept_frames = scored_frames[k_drop:]
```

`reconstruction/pipeline/run_colmap_v3.py`

PRIMARY TOOL	SUPPORTING I/O
COLMAP 4.2.0 (CPU / Multithreaded)	reconstruction/pipeline/colmap_io.py
TRACKED SCALE	**MEASURED DURATION**
370 registered cameras, 89,839 3D points	15m 12s (~12 – 18 mins)

Algorithmic Mechanism:

- **SIFT Multi-Scale Keypoints:** Extracts local scale-invariant feature descriptors across curated frames.
- **Exhaustive Matching & RANSAC:** Matches feature pairs with epipolar geometry constraint $x'^{T} E x = 0$ to filter false correspondences.
- **Non-Linear OPENCV Model:** Solves radial and tangential lens distortion coefficients alongside 3D point cloud triangulation.

```
subprocess.check_call([
    colmap_bin, "feature_extractor",
    "--database_path", db_path, "--image_path", frames_dir,
    "--ImageReader.camera_model", "OPENCV", "--ImageReader.single_camera", "1"
])

subprocess.check_call([
    colmap_bin, "exhaustive_matcher",
    "--database_path", db_path, "--FeatureMatching.use_gpu", "0"
])

subprocess.check_call([
    colmap_bin, "mapper",
    "--database_path", db_path, "--image_path", frames_dir,
    "--output_path", sparse_dir
])
```

`reconstruction/pipeline/train_v3.py`

PRIMARY TOOL	SPHERICAL HARMONICS
Brush 3DGS Engine (CUDA 13.2, RTX GPU)	Degree 2 (SH=2, 27 parameters per splat)
SCALE PENALTY	MEASURED DURATION
5e-8 SuperClean Regularization	17m 30s (~15 – 20 mins, 30k steps)

Algorithmic Mechanism:

- **3D Volumetric Representation:** Parameterizes Gaussians with position μ, scale s, rotation quaternion q, opacity α, and SH color rest coefficients.
- **SuperClean Regularization:** Penalizes large splat scales (5e-8) to eliminate cloudy floaters while retaining crisp, high-frequency room surfaces.
- **Differentiable Rasterization:** Blends photometric L1 and SSIM error over tile-based GPU rendering buffers.

```
cmd = [
    BRUSH_BIN, DATA_DIR,
    "--total-steps", "30000",
    "--growth-stop-iter", "6000",
    "--growth-grad-threshold", "0.00004",
    "--max-splats", "1000000",
    "--sh-degree", "2",
    "--export-every", "30000",
    "--export-path", export_dir,
    "--export-name", "scene.ply"
]
```

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`reconstruction/pipeline/clean_splat.py`

PRIMARY SCRIPT

clean_splat.py, sh_clamp_filter.py, filter_needles.py

FILTERING PASSES

SOR (k=16, 1.5σ) + Needle Clamping + SH Clamping

FINAL SPLAT COUNT

102,242 splats (24.3 MB)

MEASURED DURATION

1m 10s (~1 – 1.5 mins)

Algorithmic Mechanism:

- **Statistical Outlier Removal (SOR):** Queries k=16 nearest neighbors via KD-Tree; prunes splats exceeding mean + 1.5σ distance.
- **Needle Aspect Ratio Bounding:** Limits eccentricity (s_max ≤ 4.0 * s_mid) to turn spiky needles into flat surface discs.
- **SH Defect Clamping:** Simulates multi-view directions and zeroes degree-2 SH rest coefficients on splats producing dark/negative glancing radiance.

```
# Multi-view radiance validation & degree-2 zeroing
bad_splats = np.sum(np.any(total_rgb <= 0.05, axis=-1), axis=0) > 1
if np.any(bad_splats):
    for sh2 in range(9, 24):
        data[b_start:b_end, idx[f"f_rest_{sh2}"]][bad_splats] = 0.0
    for sh1 in range(0, 9):
        data[b_start:b_end, idx[f"f_rest_{sh1}"]][bad_splats] *= 0.3
```

`reconstruction/pipeline/build_scene.py`

PRIMARY TOOL	COORDINATE TARGET
Rodrigues Rotation Formula & NumPy	Three.js Standard y-up Coordinate System
ALIGNMENT QUAT	**MEASURED DURATION**
[0.931588, 0.0, -0.295638, 0.211523]	~5 seconds

Algorithmic Mechanism:

- **Camera-Up Inversion:** Calculates physical world-up vector as `V_up = -col₁(R_w2c)`.
- **Rodrigues Orientation:** Derives rotation quaternion mapping estimated floor-normal to Three.js +Y.
- **Scene Contract:** Defines player spawn `[0.2469, -0.0321, 0.871]`, eye height, and collision bounds.


```
{
  "id": "custom-video-v3D",
  "name": "TM-3D7DA1DDE3B Scene v3D (SuperClean PLY)",
  "description": "Dedicated High-Clarity Model with SuperClean scale regularization penalty.
  "coordinateSystem": "y-up",
  "transform": { "quaternion": [0.931588, 0.0, -0.295638, 0.211523] },
  "bounds": { "min": [-2.9886, -3.1241, -4.1665], "max": [3.4955, 1.705, 4.2016] },
  "spawn": { "position": [0.2469, -0.0321, 0.871], "yaw": -0.00766 },
  "assets": { "splat": "splat/scene_superclean.ply" }
}
```

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`viewer/src/main.js & viewer/vercel.json`

PRIMARY LIBRARIES	DEFAULT MODEL
Three.js 0.169 & @mkkellogg/gaussian-splats-3d 0.4.7	TM-3D7DA1DDE3B (custom-video-v3D)
CLIENT FRAME RATE	**WORKER RADIX SORT**
83 – 144+ FPS (6–12ms render time)	5 – 8ms asynchronous sort

Algorithmic Mechanism:

- **Automated Default Load:** Website loads `custom-video-v3D` immediately with zero URL parameter overhead.
- **Streamlined Controls:** UI includes only ✨ `v3D SuperClean`, 🔁 `Orbit & Zoom`, 🚶 `Walk (WASD)`, 🔭 `Angle Limit`, 🎨 `BG Mode`, and 🎯 `Center`. All deprecated prototype buttons are completely excised.
- **Cross-Origin Worker Isolation:** Configures `Cross-Origin-Opener-Policy: same-origin` and `Cross-Origin-Embedder-Policy: require-corp` for high-speed Web Worker radix sorting.

```
// viewer/src/main.js
const urlParams = new URLSearchParams(window.location.search);
const activeScene = urlParams.get('scene') || 'custom-video-v3D';
const SCENE_URL = activeScene.endsWith('.json') ? `/scenes/${activeScene}` : `/scenes/${acti

function resetToCenter() {
  setMode('orbit');
  camera.position.set(0.25, 0.2, 0.85);
  orbitControls.target.set(0.25, -0.2, -0.4);
  orbitControls.update();
}
```

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`viewer/src/poiManager.js`

PRIMARY TOOL	DETECTION TOOL
Three.js Matrix Unprojection & POIManager	scripts/poi/poi_pipeline.py (YOLO-World + Ray Cast)
VERIFIED POIS	**FLIGHT TRANSITION**
24 Room Objects (scenes/custom-video-v3D/poi_labels.json)	1200ms Cubic Ease-Out Camera Flight

Algorithmic Mechanism:

- **Frustum Screen Projection:** Converts 3D world anchors into screen coordinates every frame via `pos3D.project(camera)`.
- **Verified Indoor Catalog:** Real items identified across the room including Wardrobes, Floor/Bed Suitcases, Laptops, Office Chair, Air Conditioner, and Footwear.
- **Cubic Spline Flight:** Smoothly glides the camera directly to focused objects on click with zero jitter.

```
export class POIManager {
  update() {
    this.labels.forEach((item) => {
      const pos3D = item.worldPos.clone().project(this.camera);
      if (pos3D.z < 1.0) {
        const x = (pos3D.x * 0.5 + 0.5) * window.innerWidth;
        const y = (-pos3D.y * 0.5 + 0.5) * window.innerHeight;
        item.element.style.transform = `translate(-50%, -50%) translate3d(${x}px,${y}px,0)`;
        item.element.style.display = 'block';
      } else {
        item.element.style.display = 'none';
      }
    });
  }
}
```